Mock Midterm Examination

Machine Learning in Engineering Science

Practice candidate paper | Fall 2026 | Published 2 November 2026

This paper assesses the Week 1 to Week 10 core material through mechanisms, short calculations, code reading and evidence-based decisions. It does not require a project or execution of code.

Student name _____ Student ID _____

Practice date _____ **Duration 75 minutes** **Raw score 100 points**

Permitted resources

One A4 sheet handwritten on both sides and a standard calculator. No internet, AI assistance or execution of code. The rule concerns permitted resources, not a requirement to bring notes.

How to answer

Answer all six questions. Show enough reasoning to identify your assumptions and calculation. Fractions or sensible rounding are accepted. Write on the numbered answer sheets at the end, using labelled continuation sheets if needed. All signed residuals in this paper mean prediction minus observation. Read the stated units and deployment population.

The written paper contributes 70 percent of the midterm. The separate individual oral contributes 30 percent and lasts two minutes per student. No midterm project is required. The midterm share of the semester grade is not set by this packet.

No	Main coverage	Points	Minutes
1	Frame a vibration prediction experiment	20	14
2	Evaluate alarms under asymmetric costs	15	11
3	Relate variance retention to prediction	10	7
4	Reason about a causal sequence model	25	19
5	Diagnose variation and a changed deployment	20	14
6	Check predictions before claiming reproduction	10	7
Read first	Instructions and initial planning	0	3

Scope: Week 1 to Week 10 core, excluding holidays, optional challenges and later Agent implementation. Question 6 addresses the Week 10 workflow lesson scheduled for 9 November; return to it after that lesson if needed. This is a practice paper, not the formal examination.

This Mock uses the same assessment structure and timing, with different cases and numbers. Its oral practice appears after the written questions. The Mock has no course-grade weight.

Question 1 Frame a vibration prediction experiment

20 points | Suggested time 14 minutes

A pump dataset contains repeated runs from 18 pumps. At the start of a run, load R in percent, rotation speed U in thousands of revolutions per minute, and ambient temperature are known. Predict vibration RMS ten seconds later in mm/s. Deployment concerns previously unseen pumps. X contains only these available inputs; pump IDs are stored separately.

```
Z = StandardScaler().fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
    Z, y, test_size=0.25, random_state=12)
```

(a) 4 points State the observational unit, target, prediction time and type of supervised task.

(b) 6 points Identify two evaluation problems for unseen-pump deployment. Describe a corrected train, validation and test procedure; exact library syntax is unnecessary.

(c) 4 points A separate RAW-input model predicts RMS = 0.5 + 0.05R + 0.2U. For R = 30, U = 2 and observed RMS = 3, calculate the prediction, signed residual (prediction minus observation), absolute error and squared error.

(d) 6 points Propose a leakage-free constant baseline for RMSE, describe a fair comparison, and explain what increasing a Ridge penalty can control and what it cannot fix in part b.

Question 2 Evaluate alarms under asymmetric costs

15 points | Suggested time 11 minutes

For parts a and b, a pump alarm has been evaluated at threshold 0.7, with faulty pump as the positive class. For part c, the second table is a DIFFERENT validation set with the same class counts. A missed fault costs 8 units and a false alarm costs 1 unit. The test table reports ONLY threshold 0.7.

Test actual class	Predicted faulty	Predicted healthy
Faulty	12	8
Healthy	6	54

Validation threshold	TP	FN	FP	TN
0.4	18	2	20	40
0.7	12	8	6	54

(a) 4 points Identify TP, FN, FP and TN in the TEST table.

(b) 6 points Calculate precision, recall (sensitivity), specificity and F1. Fractions or three-significant-figure decimals are acceptable. Use F1 = 2TP / (2TP + FP + FN).

(c) 5 points Calculate the validation cost at both thresholds, choose the lower-cost threshold and explain when to freeze that decision. Can the supplied test table establish the selected threshold's final test performance?

Question 3 Relate variance retention to prediction

10 points | Suggested time 7 minutes

Two standardized pump features have covariance [[1, 0.6], [0.6, 1]]. The supplied eigenvalues are 1.6 and 0.4. Unit eigenvectors are shown below. Consider z = (2, 0). Scores are dot products; PC1-only reconstruction is score1 times v1.

$$v_1 = \frac{(1,\ 1)}{\sqrt{2}} \quad v_2 = \frac{(1,\ -1)}{\sqrt{2}}$$

(a) 2 points Calculate PC1's retained-variance fraction.

(b) 2 points Calculate z's PC1 score and reconstruction using PC1 alone.

(c) 2 points A fault is identified by the difference between the standardized features. Explain what PC1-only reconstruction can remove, even if it retains most variance.

(d) 4 points Explain how changing an unscaled coordinate from seconds to milliseconds can affect K-means, and why cluster numbers do not automatically name real fault modes.

Question 4 Reason about a causal sequence model

25 points | Suggested time 19 minutes

Use the LSTM equations below. All gate outputs and the candidate are supplied. Attention logits are already scaled, immediately before masking and softmax.

$$c_t = f_t c_{t-1} + i_t g_t$$

$$h_t = o_t \tanh(c_t)$$

(a) 4 points A series has x0 = 1, x1 = 3, x2 = 2, x3 = 5, x4 = 4. At time 3 use a four-value lookback to predict x4. Give the input and target, identify why x4 cannot enter the input, and state the split discipline for predicting a future time period.

(b) 6 points Given c_previous = 1, f_t = 0.6, i_t = 0.5, g_t = 0.8 and o_t = 0.25, calculate c_t and h_t using tanh(1) = 0.7616. Then set ONLY o_t to zero and explain what happens to the hidden output and the stored cell state.

(c) 8 points Logits are [ln(2), 0, ln(3)], with exponentials [2, 1, 3]. Key 2 is padding. V1 = [5,0], V2 = [100,100], V3 = [0,5]. Compute masked normalized weights and context. Explain why zeroing the padding weight after an ordinary unmasked softmax is insufficient.

(d) 7 points Two length-10 inputs to a fixed deterministic causal decoder share positions 0 through 3 and differ at positions 4 through 9. Which output logits must agree and why? Explain parallel teacher-forced training versus ordinary sequential autoregressive generation, and the purpose of position information.

Question 5 Diagnose variation and a changed deployment

20 points | Suggested time 14 minutes

A controlled synthetic study has independent zero-mean observation-noise variance 0.25. At one fixed input the known noise-free target is 2.0. Twenty independent training resamples give the table. Use estimated expected MSE = squared bias + refit variance + noise variance.

Model	Training rows	Mean prediction	Refit variance
Affine	80	1.6	0.25
Affine	800	1.6	0.01
Nonlinear	800	2.0	0.08

(a) 6 points Calculate squared bias and expected MSE for the 800-row affine model, and expected MSE for the 800-row nonlinear model.

(b) 4 points Explain what the larger sample changes and what it does not fix. Relate this to a persistent curved mean residual and to observation noise.

(c) 4 points Distinguish A, deployment humidity values become more frequent in a different range with the same conditional response, from B, pump ageing changes the conditional response at the same inputs.

(d) 6 points A frozen model has MAE 3 on 20 high-load cases and MAE 0.5 on 80 other test cases. Compute total test MAE and its reweighted value when high-load cases become 40 percent of deployment, assuming within-group mean absolute errors stay fixed. Explain the assumption and one limitation.

Question 6 Check predictions before claiming reproduction

10 points | Suggested time 7 minutes

A fresh receiver process refits the frozen model pair. The following row-order difference occurs for both baseline and candidate test predictions. The required input contract declares rotation speed in revolutions per minute, abbreviated rpm.

Reference ID	Prediction	Receiver ID	Prediction
9	3	7	1
4	8	9	3
7	1	4	8

(a) 3 points State the required ID checks, how to align the files, and the aligned maximum absolute prediction difference.

(b) 2 points Why is matching a rounded MAE inadequate compared with aligned vector comparison for both models at a predeclared tolerance?

(c) 2 points A copied contract changes rpm to revolutions per second while leaving values unchanged. What must an rpm-only loader do before fitting, and why?

(d) 3 points State an integrity claim supported by matching hashes and replay, two stronger claims they cannot establish, and a specific environment detail worth recording.

Written examination answer sheet

Student ID _____ Label each part a b c or d clearly.

Question 1

Question 2

Written examination answer sheet

Student ID _____ Label each part a b c or d clearly.

Question 3

Question 4

Written examination answer sheet

Student ID _____ Label each part a b c or d clearly.

Question 5

Question 6

Two minute oral practice

These two cases are practice only. They do not disclose the formal oral cards. Use 15 seconds to read, 60 seconds for the main response, 30 seconds for the follow-up and 15 seconds for closure and scoring. Practise a concise claim, the supporting evidence, and a bounded next step.

M1 Selection after test

A student inspects the test RMSE, changes Ridge alpha five times, then reports the smallest test RMSE as the final unbiased result.

Main question Identify the main evaluation problem and propose a corrected next step.

Follow up Would fixing the random seed make the final test claim valid?

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M2 A unit label in a replay

A replay package expects airflow in m/s. A copied contract says km/h while retaining the same numerical inputs. The loader rejects the contract before fitting.

Main question Is this rejection useful or a model failure? Explain the next appropriate action.

Follow up If you simply rename km/h back to m/s and obtain matching predictions, have you established that the original measurements were correctly encoded?

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